

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-0981

Joppatowne 138 kV
February 15, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0981. The Interconnection Customer proposes a 953.3 MW natural gas combined cycle facility interconnecting at the Joppatowne 138 kV substation in the Maryland area. The request was received on April 12, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 294 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$42,948,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
N STUNTON to WINDY EDGE STA 138 KV L	TANGUISSON to FULLERTON 138 KV	218.6	177	25.5
CLAREMONT to BARLING 138 KV L	FULLERTON to CLOVERLEAF 138 KV	169.8	232	26.9
FULLERTON to TANGUISSON 138 KV	N STAUNTON to WINDY EDGE STA 138 KV	169.3	250	27.6
WINDY EDGE STATION to NOTCH CLIFF SUBSTATION 138 KV	CLOVERLEAF to HAZELWOOD 138 KV line	163.2	265	24.4
JOPPATOWNE SUBSTATION to RAPHAEL RD 138 KV L	JOPPATOWNE to RAPHAEL RD 138 KV Line 2	160.9	277	26.1
TANGUISSON SUBSTATION to RAPHAEL RD 138 KV Ckt 2	TANGUISSON to RAPHAEL RD 138 KV Line	159.3	278	27.0
TANGUISSON to NORTH STAUNTON 138 KV	FULLERTON to TANGUISSON 138 KV	156.9	242	25.5
NOTCH CLIFF to CANEY CRK 138 KV	HAZELWOOD to CLOVERLEAF 138 KV line	156.6	236	24.4
TANGUISSON to RAPHAEL ROAD 138 KV	TANGUISSON to RAPHAEL ROAD 138 KV Cir. 2	154.8	286	27.0
JOPPATOWNE to RAPHAEL D 138 KV #2	RAHAEL RD to JOPPATOWNE 138 KV	142.7	313	26.1
CLOVERLEAF to FULLERTON 138 KV	WINDY EDGE STA to NOTCH CLIFF 138 KV	135.7	226	17.7
BARLING to JOPPATOWNE 138 KV Cir. 1	JOPPATOWNE SUBSTATION to BARLING 138 KV Cir. 2	129.5	312	65.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BARLING to JOPPATOWNE SUBSTATION 138 KV Circuit 2	BARLING to JOPPATOWNE 138 KV Circuit 1	128.9	313	65.4
NEWMAN to BARLING 138 KV	DODD RD 345 kV/138 kV XFMR	124.6	230	7.2
2 GENERATING STA to CLAREMONT 138 KV	CLOVERLEAF to FULLERTON 138 KV Line	119.0	258	12.1
FULLERTON to WINDY EDGE STATION 138 KV	N STAUNTON to WINDY EDGE STATION 138 KV	114.1	205	18.9
DUDEY to NEWMAN 138 KV line	DODD RD 345/138 kV Autotransformer	112.7	239	7.2
WESTERN DIGITAL to RAVENSWORTH 138 KV	HIDALGO SMELTER 345/1kV Autotransformer	109.3	301	8.7
2 GENERATING STA SUBSTATION to GUNPOWDER 138 KV	CLOVERLEAF to FULLERTON 138 KV	108.7	241	12.1
COLDSPRING to RECOVERY SOCO 138 KV	CTR to E TOWSON 138 KV line	108.7	322	7.6
BGE TMONIUM to CIPRES 138 KV	MOUNT WASHINGTON to CANEY CRK 138 KV	108.1	192	13.7
FRENCH MEADOWS SUBSTATION to RAVENSWORTH 138 KV	HIDALGO SMELTER 345/1 kV Auto	107.9	260	8.7
HAZELWOOD SUBSTATION to CLOVERLEAF 138 KV Line	HIDALGO SMELTER 345/1kV XFMR	106.4	369	8.7
E TOWSON to CENTER 138 KV	RECOVERY SOCO to COLDSPING 138 KV L	103.0	338	7.7
IBM COTTLE to DUDLEY 138 KV line	DODD ROAD 345 kV/138 kV TX	102.1	247	7.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N STUNTON to WINDY EDGE STA 138 KV line	Rebuild line with higher-capacity conductor	\$542,000	77% (\$417,000)
BARLING to CLAREMONT 138 KV Line	Rebuild line with higher-capacity conductor	\$1,842,000	93% (\$1,705,000)
FULLERTON to TANGUISSON 138 KV	Rebuild line with higher-capacity conductor	\$937,000	80% (\$752,000)
NOTCH CLIFF SUBSTATION to WINDY EDGE STA 138 KV	Rebuild line with higher-capacity conductor	\$1,803,000	81% (\$1,455,000)
JOPPATOWNE SUBSTATION to RAPHAEL RD 138 KV L	Rebuild line with higher-capacity conductor	\$2,046,000	83% (\$1,707,000)
TANGUISSON SUBSTATION to RAPHAEL RD 138 KV Ckt 2	Rebuild line with higher-capacity conductor	\$4,953,000	83% (\$4,131,000)
N STAUNTON to TANGUISSON 138 KV	Rebuild line with higher-capacity conductor	\$970,000	82% (\$790,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NOTCH CLIFF to CANEY CRK 138 KV	Rebuild line with higher-capacity conductor	\$4,158,000	81% (\$3,357,000)
TANGUISSON to RAPHAEL ROAD 138 KV	Rebuild line with higher-capacity conductor	\$6,356,000	83% (\$5,302,000)
JOPPATOWNE to RAPHAEL D 138 KV #2	Rebuild line with higher-capacity conductor	\$3,952,000	83% (\$3,297,000)
CLOVERLEAF to FULLERTON 138 KV Line	Rebuild line with higher-capacity conductor	\$2,427,000	78% (\$1,880,000)
BARLING to JOPPATOWNE 138 KV Cir. 1	Rebuild line with higher-capacity conductor	\$1,389,000	92% (\$1,282,000)
BARLING to JOPPATOWNE SUBSTATION 138 KV Circuit 2	Rebuild line with higher-capacity conductor	\$1,588,000	92% (\$1,465,000)
NEWMAN to BARLING 138 KV	Rebuild line with higher-capacity conductor	\$2,956,000	100% (\$2,956,000)
2 GENERATING STA to CLAREMONT 138 KV	Rebuild line with higher-capacity conductor	\$3,154,000	87% (\$2,742,000)
FULLERTON to WINDY EDGE STATION 138 KV	Rebuild line with higher-capacity conductor	\$2,030,000	86% (\$1,744,000)
NEWMAN to DUDLEY SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$1,530,000	100% (\$1,530,000)
RAVENSWORTH to WESTERN DIGITAL 138 KV	Reconductor line	\$572,000	70% (\$400,000)
2 GENERATING STATION to GUNPOWDER 138 KV L	Reconductor line	\$773,000	87% (\$672,000)
COLDSPRING to RECOVERY SOCO 138 KV	Reconductor line	\$1,102,000	47% (\$514,000)
BGE TMONIUM to CIPRES 138 KV	Reconductor line	\$633,000	79% (\$498,000)
FRENCH MEADOWS SUBSTATION to RAVENSWORTH 138 KV	Reconductor line	\$2,113,000	70% (\$1,477,000)
HAZELWOOD to CLOVERLEAF 138 KV	Reconductor line	\$910,000	70% (\$636,000)
E TOWSON to CENTER 138 KV	Reconductor line	\$2,011,000	55% (\$1,097,000)
IBM COTTLE to DUDLEY 138 KV line	Reconductor line	\$1,142,000	100% (\$1,142,000)
	Total allocated to project		\$42,948,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.